

Tumours of the Lung & Bronchi

The Big Picture

Lung cancer (bronchogenic carcinoma) = a top cause of cancer death worldwide, usually **diagnosed late**, overwhelmingly driven by **smoking (~80–90%)**. The master split is **NSCLC (80–85%)** vs **SCLC (10–15%)**. Know the subtypes, central vs peripheral behaviour, the spread/syndromes, **TNM staging**, and that **SCLC = chemo/radiotherapy**, **NSCLC (early) = surgery**.

Normal Anatomy (orientation)

- Right lung 3 lobes, left 2 (heart). Trachea → bronchi → bronchioles → alveoli; bronchi lined by **pseudostratified ciliated columnar epithelium**.
- Right main bronchus = wider, shorter, more vertical (aspiration). Cancers usually start in **bronchial lining cells**.

Definitions

- **Primary** (lung/bronchus) vs **metastatic** (secondary).
- **Central** (main bronchi/hilum) vs **peripheral** (outer parenchyma) — affects symptoms, imaging, biopsy route.

Aetiology

- **Smoking ~90%** of cases (active + passive +20–30%); risk highest in male smokers.
- Other: **asbestos** (dose-dependent), radon, chromium/nickel/arsenic, PAHs, **idiopathic pulmonary fibrosis**, prior chest radiation, family history.
- **Up to 20% of patients never smoked** (often adenocarcinoma).
- Central bronchi sequence: **squamous metaplasia** → **dysplasia** → **carcinoma in situ** → **invasive**.

Classification

- **NSCLC (80–85%): adenocarcinoma, squamous cell, large cell** — grouped (similar treatment/prognosis).
- **SCLC (10–15%):** aggressive neuroendocrine, "oat cell."

Adenocarcinoma

- **Most common type now**; more in **non-smokers, women, younger**; **peripheral**, gland-forming.
- Often asymptomatic early (peripheral); ground-glass opacities; spreads to **brain, liver, bone, adrenals**.

Squamous Cell Carcinoma

- **Central**, near main bronchus; **smoking-linked**.
- Cough, **haemoptysis**, recurrent pneumonia (obstruction), **cavitation**.

Large Cell (+ LCNEC)

- Anywhere, grows/spreads fast; LCNEC behaves like SCLC.

SCLC

- Central, almost always smokers; **early metastasis** (usually spread at diagnosis); responds to **chemo + radiotherapy** but relapses.

Other Tumours

- **Carcinoid** (<5%, slow, may cause **carcinoid syndrome**), atypical carcinoid, **pulmonary hamartoma** (commonest benign, peripheral "**coin lesion**" — cartilage/fat), sarcoma, **MALT lymphoma**, metastases ("**cannonball**" nodules from breast/colon/kidney/testis/sarcoma).

Spread & Syndromes

- **Local:** pleura, chest wall, mediastinum, great vessels, oesophagus → effusion, **SVC obstruction**.
- **Lymphatic:** peribronchial → hilar → mediastinal → supraclavicular (the N in TNM).
- **Distant: brain, bone, liver, adrenals** (= stage IV).
- Syndromes: **SVC syndrome** (face/neck/arm swelling), **Horner's** (small pupil, ptosis, anhidrosis), hoarseness, shoulder pain.

Diagnosis & Staging

- History/exam → **CXR** (mass, collapse, effusion) → **CT** → **biopsy**; **PET-CT** for spread.
- **Bronchoscopy** (central tumours: visualise + biopsy/BAL), **CT-guided needle** (peripheral), **EBUS** (nodes), VATS.
- **IHC markers:** TTF-1/Napsin A → adeno; **p40/p63** → **squamous**; chromogranin/synaptophysin/CD56 → neuroendocrine.
- **TNM:** T (size/local invasion), N (N0→N3 nodes), M (M0/M1; M1a pleural/contralateral, M1b single distant, M1c multiple). Stages I–IV.

Management

- **NSCLC:** early → **surgery (lobectomy standard;** segment/wedge selected; pneumonectomy extensive) + adequate cardiopulmonary reserve; later → chemo/radiotherapy ± targeted/immune therapy.
- **SCLC:** mainly **chemo (platinum) + radiotherapy**.
- **Radiotherapy:** curative SBRT (localised) or palliative (bone/brain).
- **Prevention:** tobacco control; **low-dose CT screening** for high-risk smokers.

Walk-Away Points

- **Smoking ~90%;** NSCLC 80–85% (adeno = peripheral/non-smoker/women; squamous = central/cavitating), SCLC = aggressive central, early spread.
- **Adenocarcinoma = most common type now;** hamartoma = benign coin lesion.
- Mets go to **brain, bone, liver, adrenals;** watch **SVC + Horner's** syndromes.
- **NSCLC early = lobectomy; SCLC = chemo + radiotherapy.**

Tumours of Lung & Bronchi — Revision Table		
Group	Share	Behaviour
NSCLC	80–85%	Slower, often surgical
SCLC	10–15%	Aggressive, early mets, chemo/RT
NSCLC subtype	Location	Notes
Adenocarcinoma	Peripheral	Commonest , non-smokers, women, younger
Squamous cell	Central	Smoking, cavitation, haemoptysis
Large cell / LCNEC	Any	Fast, LCNEC ~ SCLC
Cause	Detail	
Smoking	~ 90% of cases	
Other	Asbestos, radon, IPF, radiation	
Never-smokers	Up to 20% (adeno)	
Other tumour	Note	
Carcinoid	<5%, carcinoid syndrome	
Hamartoma	Commonest benign, coin lesion	
Metastasis	"Cannonball" nodules	
Spread	Detail	
Distant sites	Brain, bone, liver, adrenals	
Local syndromes	SVC obstruction, Horner's , hoarseness	
Nodes	Hilar → mediastinal → supraclavicular	
IHC marker	Tumour	
TTF-1 / Napsin A	Adenocarcinoma	
p40 / p63	Squamous	
Chromogranin/synaptophysin/CD56	Neuroendocrine	
TNM	Component	
T	Size + local invasion	
N	N0–N3 nodes	
M	M1a/b/c distant	
Management	Detail	
NSCLC early	Lobectomy	
SCLC	Chemo + radiotherapy	
Screening	Low-dose CT (high-risk smokers)	

Bold = high-yield.